

**Laboratorium  
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# **Laporan Akhir Praktikum Jaringan Komputer**

## **Vlan Trunk OSPF MultiVendor**

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# 1 Langkah-Langkah Percobaan

## 1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Membuka terminal Cisco Switch kemudian masuk ke mode konfigurasi untuk melakukan pembuatan VLAN dan konfigurasi trunk sesuai topologi praktikum.
2. Membuat VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 pada Cisco Switch serta mengkonfigurasi access port sesuai kebutuhan jaringan.
3. Mengkonfigurasi interface yang terhubung ke Cisco Router sebagai trunk port sehingga dapat meneruskan seluruh VLAN yang digunakan.

```
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status        Native vlan
Gi0/0     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     none
```

**Gambar 1:** Verifikasi Konfigurasi Trunk Port

4. Melakukan verifikasi VLAN menggunakan perintah `show vlan brief` untuk memastikan seluruh VLAN telah berhasil dibuat.

```
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
                                           Gi1/0, Gi1/1, Gi1/2, Gi1/3
10   Finance                 active
20   HR                      active
30   IT                       active
40   Marketing                active
1002 fddi-default           act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
```

**Gambar 2:** Verifikasi VLAN pada Cisco Switch

5. Menyimpan konfigurasi yang telah dibuat menggunakan perintah `write memory`.

```
Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1652 bytes[OK]
Switch#
*Jun  5 08:38:52.282: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 08:38:52.982: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
```

**Gambar 3:** Penyimpanan Konfigurasi Cisco Switch

## 1.2 Praktikum 2: Konfigurasi Cisco Router

1. Membuka terminal Cisco Router kemudian mengkonfigurasi subinterface untuk VLAN 10, VLAN 20, VLAN 30, dan VLAN 40 sebagai gateway masing-masing jaringan.

2. Mengkonfigurasi DHCP Server pada VLAN 10 dan VLAN 20 sesuai dengan kebutuhan jaringan.
3. Melakukan verifikasi interface menggunakan perintah `show ip interface brief`.

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  up                    up
GigabitEthernet0/2.10 192.168.10.1   YES manual  up                    up
GigabitEthernet0/2.20 192.168.20.1   YES manual  up                    up
GigabitEthernet0/2.30 192.168.30.1   YES manual  up                    up
GigabitEthernet0/2.40 192.168.40.1   YES manual  up                    up
GigabitEthernet0/3 unassigned      YES unset  administratively down down
GigabitEthernet0/4 unassigned      YES unset  administratively down down
```

**Gambar 4:** Verifikasi Interface Cisco Router

4. Melakukan verifikasi DHCP Pool menggunakan perintah `show ip dhcp pool`.

```
Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index   IP address range   Leased addresses
192.168.10.12   192.168.10.1 - 192.168.10.254   1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index   IP address range   Leased addresses
192.168.20.12   192.168.20.1 - 192.168.20.254   1
```

**Gambar 5:** Verifikasi DHCP Pool

5. Melakukan verifikasi DHCP Binding menggunakan perintah `show ip dhcp binding`.

```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.6e Jun 06 2026 08:51 AM    Automatic
192.168.20.11    0100.5079.6668.6d Jun 06 2026 08:51 AM    Automatic
```

**Gambar 6:** Verifikasi DHCP Binding

6. Melakukan verifikasi alamat IP yang diperoleh PC pada VLAN 10.

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86392, 86400/43200/75600
MAC        : 00:50:79:66:68:6e
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500
```

**Gambar 7:** Verifikasi IP Client VLAN 10

7. Melakukan verifikasi alamat IP yang diperoleh PC pada VLAN 20.

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86398, 86400/43200/75600
MAC        : 00:50:79:66:68:6d
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500
```

**Gambar 8:** Verifikasi IP Client VLAN 20

8. Melakukan pengujian konektivitas menuju gateway VLAN 10.

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.208 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.336 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.133 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.763 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.018 ms
```

**Gambar 9:** Pengujian Gateway VLAN 10

9. Melakukan pengujian konektivitas menuju gateway VLAN 20.

```
VPCS> ping 192.168.20.1  
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=3.000 ms  
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.293 ms  
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.633 ms  
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.136 ms  
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=2.776 ms
```

**Gambar 10:** Pengujian Gateway VLAN 20

10. Melakukan pengujian konektivitas menuju gateway VLAN 30.

```
VPCS> ping 192.168.30.1  
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.925 ms  
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.565 ms  
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.035 ms  
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=5.439 ms  
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.268 ms
```

**Gambar 11:** Pengujian Gateway VLAN 30

11. Melakukan pengujian konektivitas menuju gateway VLAN 40.

```
VPCS> ping 192.168.40.1  
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=3.703 ms  
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.404 ms  
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.323 ms  
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.380 ms  
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=3.613 ms
```

**Gambar 12:** Pengujian Gateway VLAN 40

12. Melakukan pengujian komunikasi antar VLAN menggunakan perintah `ping` dari VLAN 10 menuju VLAN lainnya.

```

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=5.048 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=4.911 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.151 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=5.608 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=4.899 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=4.810 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.967 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.851 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.641 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=5.020 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=4.762 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=3.950 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.696 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=5.910 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=4.102 ms

```

**Gambar 13:** Pengujian Inter-VLAN Routing

### 1.3 Praktikum 3: Konfigurasi Huawei Router dan MikroTik

1. Membuka terminal Huawei Router kemudian mengkonfigurasi alamat IP pada interface yang terhubung ke jaringan lokal dan jaringan antar router.
2. Melakukan verifikasi konfigurasi interface pada Huawei Router menggunakan perintah `display ip interface brief`.

```

[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask      Physical  Protocol VPN
Ethernet1/0/0             10.10.10.2/30        up        up      --
Ethernet1/0/0.4094        128.1.138.137/16    up        up      l3vpn
Ethernet1/0/1             unassigned            up        down    --
Ethernet1/0/1.4094        128.1.138.137/16    up        up      l3vpn
Ethernet1/0/2             192.168.50.1/24      up        up      --
Ethernet1/0/2.4094        128.1.138.137/16    up        up      l3vpn
Ethernet1/0/3             10.10.20.1/30        up        up      --
Ethernet1/0/3.4094        128.1.138.137/16    up        up      l3vpn
Ethernet1/0/4             10.10.11.2/30        up        up      --
Ethernet1/0/4.4094        128.1.138.137/16    up        up      l3vpn
---- More ----

```

**Gambar 14:** Verifikasi Interface Huawei Router

3. Melakukan verifikasi status interface pada Cisco Router menggunakan perintah `show ip interface brief`.

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.10.11.1      YES manual up          up
GigabitEthernet0/1       10.10.30.1      YES manual up          up
GigabitEthernet0/2       unassigned      YES unset up          up
GigabitEthernet0/2.10    192.168.10.1    YES manual up          up
GigabitEthernet0/2.20    192.168.20.1    YES manual up          up
GigabitEthernet0/2.30    192.168.30.1    YES manual up          up
GigabitEthernet0/2.40    192.168.40.1    YES manual up          up
GigabitEthernet0/3       10.10.10.1      YES manual up          up
GigabitEthernet0/4       unassigned      YES unset administratively down down
```

**Gambar 15:** Verifikasi Interface Cisco Router

4. Mengkonfigurasi alamat IP pada MikroTik sesuai dengan topologi jaringan.

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                SIZE TTL TIME   STATUS
0 10.10.20.1            56 255 3ms
1 10.10.20.1            56 255 1ms
2 10.10.20.1            56 255 1ms
3 10.10.20.1            56 255 1ms
4 10.10.20.1            56 255 1ms
5 10.10.20.1            56 255 2ms
6 10.10.20.1            56 255 1ms
7 10.10.20.1            56 255 1ms
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                SIZE TTL TIME   STATUS
0 10.10.30.1            56 255 2ms
1 10.10.30.1            56 255 1ms
2 10.10.30.1            56 255 4ms
3 10.10.30.1            56 255 6ms
4 10.10.30.1            56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=6ms
```

**Gambar 16:** Konfigurasi IP Address MikroTik

5. Melakukan verifikasi konfigurasi IP pada router menggunakan perintah `show ip route`.

```
VPCS> show ip

NAME           : VPCS[1]
IP/MASK        : 192.168.50.10/24
GATEWAY        : 192.168.50.1
DNS            :
MAC           : 00:50:79:66:68:76
LPORT         : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU           : 1500
```

**Gambar 17:** Verifikasi Konfigurasi Routing

6. Melakukan pengujian konektivitas menuju gateway jaringan.

```
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=9.810 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.739 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=4.161 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.166 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.408 ms
```

**Gambar 18:** Pengujian Gateway

7. Melakukan pengujian komunikasi antar router menggunakan perintah ping.

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/3 ms
Router#
```

**Gambar 19:** Pengujian Antar Router

8. Melakukan pengujian konektivitas dari MikroTik menuju Huawei Router.

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                            56 255 3ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 1ms
4 10.10.20.1                            56 255 1ms
5 10.10.20.1                            56 255 2ms
6 10.10.20.1                            56 255 1ms
7 10.10.20.1                            56 255 1ms
sent=8 received=8 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                            56 255 2ms
1 10.10.30.1                            56 255 1ms
2 10.10.30.1                            56 255 4ms
3 10.10.30.1                            56 255 6ms
4 10.10.30.1                            56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=6ms
```

**Gambar 20:** Pengujian Konektivitas MikroTik ke Huawei

## 1.4 Praktikum 4: Konfigurasi OSPF Multi Vendor

1. Membuka terminal Cisco Router kemudian mengaktifkan protokol routing OSPF dan menambahkan seluruh network yang terhubung ke router ke dalam area OSPF sesuai topologi jaringan.
2. Melakukan verifikasi konfigurasi OSPF pada Cisco Router menggunakan perintah `show ip protocols`.



```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

```

**Gambar 21:** Verifikasi Konfigurasi OSPF pada Cisco Router

3. Melakukan verifikasi neighbor OSPF pada Cisco Router menggunakan perintah `show ip ospf neighbor` untuk memastikan hubungan adjacency telah terbentuk.

```

Router#show ip ospf neighbor

```

| Neighbor ID | Pri | State    | Dead Time | Address    | Interface          |
|-------------|-----|----------|-----------|------------|--------------------|
| 3.3.3.3     | 1   | FULL/BDR | 00:00:37  | 10.10.30.2 | GigabitEthernet0/1 |
| 2.2.2.2     | 1   | FULL/BDR | 00:00:35  | 10.10.11.2 | GigabitEthernet0/0 |
| 2.2.2.2     | 1   | FULL/BDR | 00:00:32  | 10.10.10.2 | GigabitEthernet0/3 |

**Gambar 22:** Verifikasi Neighbor OSPF Cisco

4. Melakukan verifikasi routing table OSPF pada Cisco Router menggunakan perintah `show ip route ospf`.

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:46 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:06:46 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:06:46 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:02:49, GigabitEthernet0/0
          [110/110] via 10.10.10.2, 00:02:49, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:07:02, GigabitEthernet0/0
          [110/11] via 10.10.10.2, 00:07:02, GigabitEthernet0/3

```

**Gambar 23:** Verifikasi Routing OSPF Cisco

5. Membuka terminal Huawei Router kemudian mengaktifkan protokol routing OSPF dan menambahkan seluruh network yang terhubung ke dalam area OSPF.

6. Melakukan verifikasi neighbor OSPF pada Huawei Router menggunakan perintah `display ospf peer`.

```
[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

          OSPF Process 1 with Router ID 2.2.2.2
          Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.10.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1          BDR: 10.10.10.2          MTU: 1500
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00h05m41s
  Neighbor Up Time : 2026-06-05 09:27:36
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3          Address: 10.10.20.2
  State: Full              Mode:Nbr is Master     Priority: 1
  DR: 10.10.20.1          BDR: 10.10.20.2          MTU: 1500
  Dead timer due in 33 sec
  Retrans timer interval: 5
  Neighbor is up for 00h01m36s
  Neighbor Up Time : 2026-06-05 09:31:42
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.11.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1          BDR: 10.10.11.2          MTU: 1500
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00h05m41s
  Neighbor Up Time : 2026-06-05 09:27:36
  Authentication Sequence: [ 0 ]
```

**Gambar 24:** Verifikasi Neighbor OSPF Huawei

7. Melakukan verifikasi routing table OSPF pada Huawei Router menggunakan perintah `display ip routing-table protocol ospf`.

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
10.10.30.0/30       OSPF    10   110        D   10.10.11.1       Ethernet1/0/4
10.10.30.0/30       OSPF    10   110        D   10.10.10.1       Ethernet1/0/0
192.168.10.0/24      OSPF    10   11          D   10.10.11.1       Ethernet1/0/4
192.168.10.0/24      OSPF    10   11          D   10.10.10.1       Ethernet1/0/0
192.168.20.0/24      OSPF    10   11          D   10.10.11.1       Ethernet1/0/4
192.168.20.0/24      OSPF    10   11          D   10.10.10.1       Ethernet1/0/0
192.168.30.0/24      OSPF    10   11          D   10.10.11.1       Ethernet1/0/4
192.168.30.0/24      OSPF    10   11          D   10.10.10.1       Ethernet1/0/0
192.168.40.0/24      OSPF    10   11          D   10.10.11.1       Ethernet1/0/4
192.168.40.0/24      OSPF    10   11          D   10.10.10.1       Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
10.10.10.0/30        OSPF    10   10          10.10.10.2       Ethernet1/0/0
10.10.11.0/30        OSPF    10   10          10.10.11.2       Ethernet1/0/4
10.10.20.0/30        OSPF    10  100          10.10.20.1       Ethernet1/0/3
192.168.50.0/24      OSPF    10   1           192.168.50.1     Ethernet1/0/2
```

**Gambar 25:** Verifikasi Routing Table OSPF Huawei

8. Melakukan verifikasi rute yang dipelajari melalui OSPF pada Huawei Router menggunakan perintah `display ospf routing`.

```
[~HUAWEI]display ospf routing
```

OSPF Process 1 with Router ID 2.2.2.2  
Routing Tables

| Routing for Network Destination | Cost | Type   | NextHop      | AdvRouter | Area    |
|---------------------------------|------|--------|--------------|-----------|---------|
| 10.10.10.0/30                   | 10   | Direct | 10.10.10.2   | 2.2.2.2   | 0.0.0.0 |
| 10.10.11.0/30                   | 10   | Direct | 10.10.11.2   | 2.2.2.2   | 0.0.0.0 |
| 10.10.20.0/30                   | 100  | Direct | 10.10.20.1   | 2.2.2.2   | 0.0.0.0 |
| 10.10.30.0/30                   | 110  | Stub   | 10.10.11.1   | 1.1.1.1   | 0.0.0.0 |
| 10.10.30.0/30                   | 110  | Stub   | 10.10.10.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.10.0/24                 | 11   | Stub   | 10.10.11.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.10.0/24                 | 11   | Stub   | 10.10.10.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.20.0/24                 | 11   | Stub   | 10.10.11.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.20.0/24                 | 11   | Stub   | 10.10.10.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.30.0/24                 | 11   | Stub   | 10.10.11.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.30.0/24                 | 11   | Stub   | 10.10.10.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.40.0/24                 | 11   | Stub   | 10.10.11.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.40.0/24                 | 11   | Stub   | 10.10.10.1   | 1.1.1.1   | 0.0.0.0 |
| 192.168.50.0/24                 | 1    | Direct | 192.168.50.1 | 2.2.2.2   | 0.0.0.0 |

Total Nets: 9  
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137  
Routing Tables

| Routing for Network Destination | Cost | Type   | NextHop       | AdvRouter     | Area    |
|---------------------------------|------|--------|---------------|---------------|---------|
| 128.1.138.137/32                | 0    | Direct | 128.1.138.137 | 128.1.138.137 | 0.0.0.0 |

Total Nets: 1  
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0

**Gambar 26:** Verifikasi OSPF Routing Huawei

9. Membuka Winbox MikroTik kemudian mengaktifkan OSPF dan menambahkan network yang diperlukan sesuai topologi jaringan.
10. Melakukan verifikasi neighbor OSPF pada MikroTik menggunakan perintah `/routing ospf neighbor print`.

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m13s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m9s
```

**Gambar 27:** Verifikasi Neighbor OSPF MikroTik

11. Melakukan verifikasi route yang diperoleh dari OSPF pada MikroTik menggunakan perintah `/ip route print where ospf`.

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.30.1 110
    10.10.20.1
1 ADo 10.10.11.0/30      10.10.30.1 110
    10.10.20.1
2 ADo 192.168.10.0/24     10.10.30.1 110
3 ADo 192.168.20.0/24     10.10.30.1 110
4 ADo 192.168.30.0/24     10.10.30.1 110
5 ADo 192.168.40.0/24     10.10.30.1 110
6 ADo 192.168.50.0/24     10.10.20.1 110
```

**Gambar 28:** Verifikasi Route OSPF MikroTik

12. Melakukan pengujian konektivitas end-to-end dari PC pada jaringan LAN Huawei menuju alamat IP VLAN pada jaringan Cisco menggunakan perintah ping.

```
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=10.664 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=6.840 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=5.232 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.193 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=5.597 ms

VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=14.587 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=6.830 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=5.803 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=6.150 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=8.657 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=11.039 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=7.986 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.669 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=7.409 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=5.744 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=5.595 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=6.390 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=7.924 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=5.676 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=5.837 ms
```

**Gambar 29:** Pengujian Konektivitas End-to-End

## 1.5 Praktikum 5: Pengujian Failover OSPF

1. Memastikan kedua jalur koneksi antara Cisco Router dan Huawei Router berada dalam kondisi aktif sebelum dilakukan pengujian failover.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.377 ms 4.617 ms 5.323 ms
 2 10.10.10.2 5.596 ms 4.418 ms 5.219 ms
 3 *192.168.50.10 9.551 ms (ICMP type:3, code:3, Destination port unreachable)
```

**Gambar 30:** Kondisi Normal Dua Link Aktif

2. Menonaktifkan salah satu link antara Cisco Router dan Huawei Router untuk mengamati kemampuan OSPF dalam melakukan perpindahan jalur secara otomatis.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.326 ms 5.060 ms 3.293 ms
 2 10.10.11.2 6.736 ms 4.479 ms 4.193 ms
 3 *192.168.50.10 5.529 ms (ICMP type:3, code:3, Destination port unreachable)
```

**Gambar 31:** Pengujian Failover Satu Link

3. Menonaktifkan kedua link utama antara Cisco Router dan Huawei Router sesuai skenario pengujian.

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.090 ms 2.903 ms 3.369 ms
 2 10.10.30.2 6.229 ms 5.606 ms 2.770 ms
 3 10.10.20.1 8.770 ms 5.739 ms 6.296 ms
 4 *192.168.50.10 11.768 ms (ICMP type:3, code:3, Destination port unreachable)
```

**Gambar 32:** Pengujian Failover Dua Link

## 2 Analisis Hasil Percobaan

Berdasarkan seluruh percobaan yang telah dilakukan, konfigurasi jaringan pada perangkat Cisco Switch, Cisco Router, Huawei Router, dan MikroTik Router berhasil diterapkan sesuai dengan topologi yang telah dirancang. Pada tahap awal, VLAN berhasil dibuat pada Cisco Switch dan interface trunk dapat berfungsi dengan baik sehingga beberapa VLAN dapat dilewatkan melalui satu jalur koneksi menuju router. Selanjutnya, konfigurasi Inter-VLAN Routing pada Cisco Router berjalan dengan baik yang ditunjukkan oleh keberhasilan setiap VLAN dalam mengakses gateway masing-masing serta kemampuan perangkat pada VLAN yang berbeda untuk saling berkomunikasi. Konfigurasi DHCP juga berhasil memberikan alamat IP secara otomatis kepada client sesuai dengan rentang alamat yang telah ditentukan. Pada praktikum berikutnya, konfigurasi alamat IP pada Huawei Router dan MikroTik berhasil dilakukan sehingga seluruh perangkat dapat saling terhubung dan berkomunikasi melalui pengujian ping antar router maupun ke gateway jaringan. Setelah konektivitas dasar berhasil dibangun, implementasi protokol routing OSPF pada ketiga perangkat menunjukkan hasil yang sesuai, ditandai dengan terbentuknya hubungan neighbor OSPF dan munculnya rute-rute baru pada routing table masing-masing perangkat. Hal ini memungkinkan jaringan yang berbeda segmen dapat saling bertukar informasi routing secara otomatis. Pengujian konektivitas dari jaringan Huawei menuju jaringan VLAN pada Cisco juga berhasil dilakukan tanpa kendala. Pada tahap akhir, dilakukan pengujian failover dengan memutus salah satu jalur koneksi antar router. Hasil pengujian menunjukkan bahwa OSPF mampu melakukan perpindahan jalur secara otomatis ke link alternatif yang masih aktif sehingga komunikasi tetap dapat berlangsung. Ketika jalur utama diaktifkan kembali, routing kembali menyesuaikan secara otomatis.

### 3 Hasil Tugas Modul

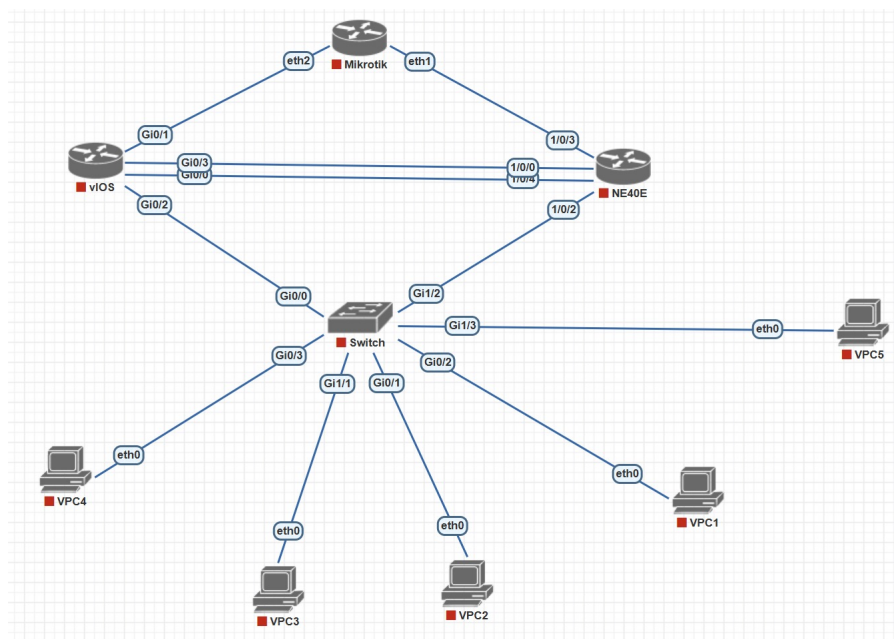
Hasil tugas modul dapat dilihat di READE.MD pada github kami

### 4 Kesimpulan

Berdasarkan praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi VLAN, trunk, Inter-VLAN Routing, DHCP, serta routing dinamis OSPF pada perangkat Cisco, Huawei, dan MikroTik berhasil diterapkan dengan baik. Seluruh perangkat dapat saling terhubung sesuai dengan topologi yang digunakan, baik dalam satu jaringan maupun antar jaringan yang berbeda. Selain itu, pengujian failover menunjukkan bahwa OSPF mampu menyesuaikan jalur komunikasi secara otomatis ketika terjadi gangguan pada salah satu link sehingga konektivitas jaringan tetap terjaga. Dengan demikian, praktikum ini memberikan pemahaman mengenai penerapan konfigurasi jaringan multi-vendor serta penggunaan routing dinamis untuk meningkatkan fleksibilitas dan keandalan jaringan.

### 5 Lampiran

#### 5.1 Dokumentasi saat praktikum



**Gambar 33:** Dokumentasi Topologi Saat Praktikum